Sentinel Fraud Detection Project

Step-by-step run instructions for the React frontend and Python fraud prediction backend.

1. Install project dependencies

Open a terminal in the project root folder and install JavaScript packages for the frontend app:

```
cd clarity-fraud-watch-main
```

```
npm install
```

Then install required Python packages for running the predictor scripts:

```
cd ..
```

```
python -m pip install numpy joblib scikit-learn reportlab
```

2. Start the React frontend

From the frontend folder start the development server and open the browser when ready:

```
cd clarity-fraud-watch-main
```

```
npm run dev
```

Open the app in your browser at the address shown by Vite (typically **http://localhost:5173**).

3. Use the transaction prediction page

In the app, navigate to **/app/predictions**. Enter transaction metadata and features, then click Predict fraud.

You can also upload a batch JSON file with multiple transactions for bulk scoring.

4. Example sample input files

The workspace includes these sample input files:

File	Description
sample_transaction.json	A realistic single transaction payload for predictive scoring.
sample_batch.json	A small batch of transactions for bulk prediction testing.

test_batch.csv	CSV payload matching the creditcard.csv feature layout for batch upload tests.

5. Run the Python prediction scripts manually

Use the sample JSON file to test the Python inference pipeline outside the UI:

```
python predict.py sample_transaction.json
```

For batch scoring with the JSON batch file:

```
python predict_batch.py < sample_batch.json
```

If you want to run CSV batch scoring directly from the frontend, use the app upload control on the Predictions page.

6. Confirm runtime files are present

Make sure these files exist in the workspace root before running:

Filename	Required for
model.joblib	Saved fraud model weights used by predict.py
scaler.joblib	Feature scaler used before prediction
creditcard.csv	Original dataset used for training and verification
test_batch.csv	Batch upload test CSV for the app prediction feature

7. Troubleshooting

If the frontend does not load, verify that the development server is running and the terminal shows no errors.

If the Python scripts fail, ensure the correct Python interpreter is used and that joblib/numpy are installed.